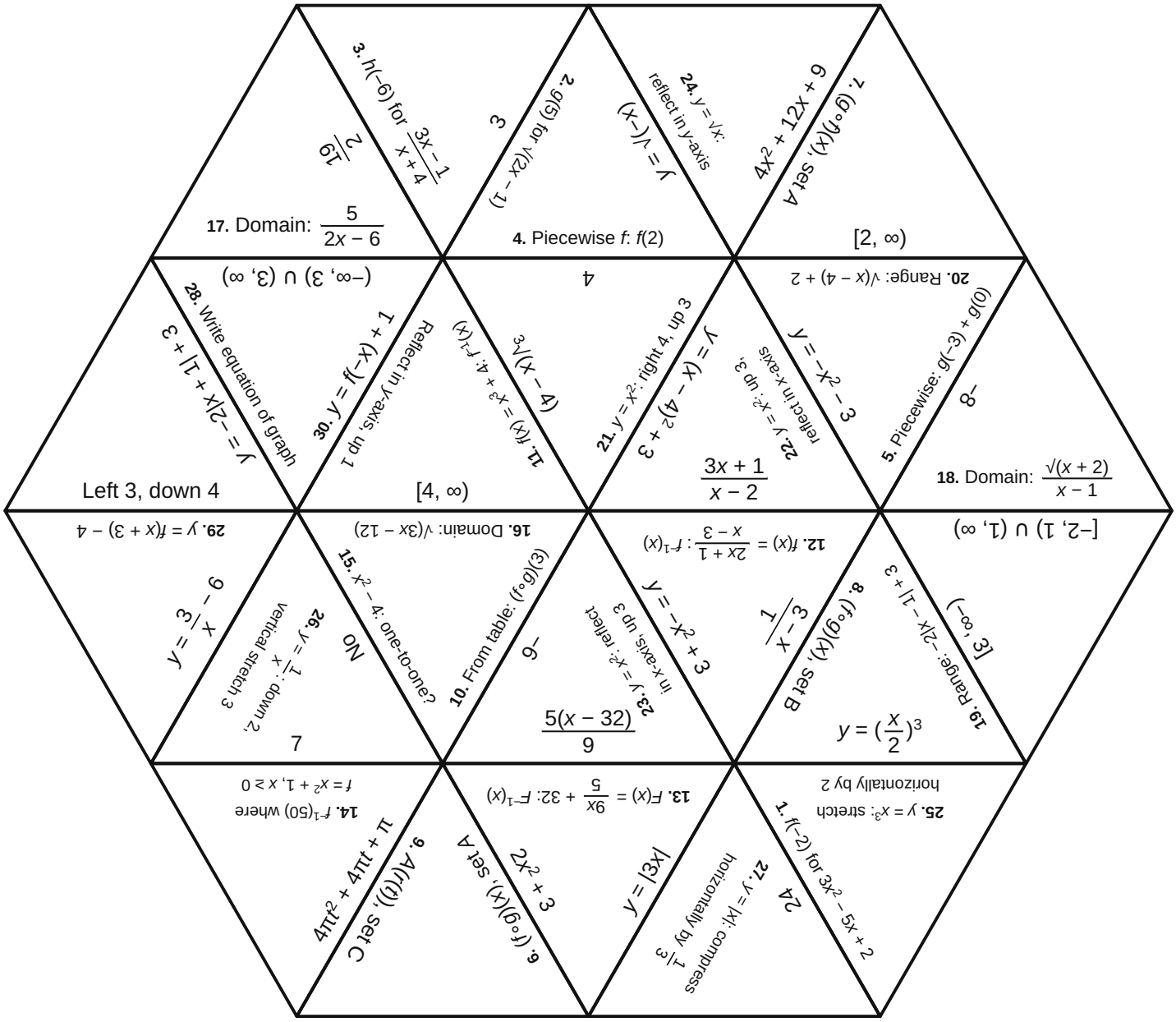
Functions & Transformations Tarsia: Answer Key

Solved hexagon. Each shared edge pairs a problem with its answer.



Answer List

Functions & Transformations Tarsia · all 30 matches

#	Problem	Answer (on the matching edge)
1	$f(-2)$ for $3x^2 - 5x + 2$	24
2	$g(5)$ for $\sqrt{(2x - 1)}$	3
3	$h(-6)$ for $\frac{3x - 1}{x + 4}$	$\frac{19}{2}$
4	Piecewise f: $f(2)$	4
5	Piecewise: $g(-3) + g(0)$	-8
6	$(f \circ g)(x)$, set A	$2x^2 + 3$
7	$(g \circ f)(x)$, set A	$4x^2 + 12x + 9$
8	$(f \circ g)(x)$, set B	$\frac{1}{x - 3}$
9	$A(r(t))$, set C	$4\pi t^2 + 4\pi t + \pi$
10	From table: $(f \circ g)(3)$	-6
11	$f(x) = x^3 + 4$: $f^{-1}(x)$	$\sqrt[3]{(x - 4)}$
12	$f(x) = \frac{2x + 1}{x - 3}$: $f^{-1}(x)$	$\frac{3x + 1}{x - 2}$
13	$F(x) = \frac{9x}{5} + 32$: $F^{-1}(x)$	$\frac{5(x - 32)}{9}$
14	$f^{-1}(50)$ where $f = x^2 + 1$, $x \geq 0$	7
15	$x^2 - 4$: one-to-one?	No
16	Domain: $\sqrt{(3x - 12)}$	$[4, \infty)$
17	Domain: $\frac{5}{2x - 6}$	$(-\infty, 3) \cup (3, \infty)$
18	Domain: $\frac{\sqrt{(x + 2)}}{x - 1}$	$[-2, 1) \cup (1, \infty)$
19	Range: $-2\lvert x - 1\rvert + 3$	$(-\infty, 3]$
20	Range: $\sqrt{(x - 4)} + 2$	$[2, \infty)$
21	$y = x^2$: right 4, up 3	$y = (x - 4)^2 + 3$
22	$y = x^2$: up 3, reflect in x-axis	$y = -x^2 - 3$
23	$y = x^2$: reflect in x-axis, up 3	$y = -x^2 + 3$
24	$y = \sqrt{x}$: reflect in y-axis	$y = \sqrt{(-x)}$
25	$y = x^3$: stretch horizontally by 2	$y = (\frac{x}{2})^3$
26	$y = \frac{1}{x}$: down 2, vertical stretch 3	$y = \frac{3}{x} - 6$
27	$y = \lvert x\rvert$: compress horizontally by $\frac{1}{3}$	$y = \lvert 3x\rvert$
28	Write equation of graph	$y = -2\lvert x + 1\rvert + 3$
29	$y = f(x + 3) - 4$	Left 3, down 4
30	$y = f(-x) + 1$	Reflect in y-axis, up 1